UTS UNIVERSITY OF TECHNOLOGY SYDNEY

**Faculty of Engineering and IT**

# Assignment Cover Sheet

**Subject:** 41277 Control Design

**Assignment Number:** Demonstration Practical 2 **Date Submitted:** [TODO: dd/mm/2026]

**Assignment Title:** State-Feedback Control

**Student Name(s) and Number(s)** **Tutorial Group:** 02


| | |
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**Signature** J.A. | E.K. | R.W. **Date** [TODO: date]

**Statement of Collaboration:**

[TODO: Describe how the team collaborated on Q1–Q3.]

**Demonstration Practice 2**

| Student Number | Student Name | Overall Contribution Weighting From 0 to 100% |
|---|---|---|
| 24767939 | Jaylen Avtarovski | 100% |
| 24817532 | Rhyse Williams | 100% |
| 24949885 | Elina Kim | 100% |

Each member's individual mark is the group mark scaled by their contribution level. If not indicated, all students get the same mark.

## Answers to the Assignment Questions:

## Q1: State-Space Model (40 marks)

### Q1.1) System Modelling

Based on [1], the QUBE-Servo 2 is modelled as

$$\dot{x}(t) = A_c\, x(t) + B_c\, u(t), \tag{1a}$$
$$y(t) = C_c\, x(t) + D_c\, u(t), \tag{1b}$$

with $n =$ **[TODO:$n$]** states, $m =$ **[TODO:$m$]** inputs and $p =$ **[TODO:$p$]** outputs.

**[TODO:** Derivation of the motor/load dynamics and the state vector $x(t)$.**]**

### Q1.2) System Parameters

The physical parameters of the QUBE-Servo 2 are given in Table 1, taken from the manufacturer's user manual [3].

Table 1: QUBE-Servo 2 system parameters [3].

| Symbol | Description | Value |
|---|---|---|
| *DC motor* | | |
| $R_m$ | Terminal resistance | $8.4\,\Omega$ |
| $k_t$ | Torque constant | $0.042\,\mathrm{N\,m/A}$ |
| $k_m$ | Motor back-emf constant | $0.042\,\mathrm{V/(rad/s)}$ |
| $J_m$ | Rotor inertia | $4.0 \times 10^{-6}\,\mathrm{kg\,m^2}$ |
| $L_m$ | Rotor inductance | $1.16\,\mathrm{mH}$ |
| $m_h$ | Module attachment hub mass | $0.0106\,\mathrm{kg}$ |
| $r_h$ | Module attachment hub radius | $0.0111\,\mathrm{m}$ |
| $J_h$ | Module attachment hub inertia | $0.6 \times 10^{-6}\,\mathrm{kg\,m^2}$ |
| *Inertia disc module* | | |
| $m_d$ | Disc mass | $0.053\,\mathrm{kg}$ |
| $r_d$ | Disc radius | $0.0248\,\mathrm{m}$ |
| *Rotary pendulum module* | | |
| $m_r$ | Rotary arm mass | $0.095\,\mathrm{kg}$ |
| $L_r$ | Rotary arm length | $0.085\,\mathrm{m}$ |
| $m_p$ | Pendulum link mass | $0.024\,\mathrm{kg}$ |
| $L_p$ | Pendulum link length | $0.129\,\mathrm{m}$ |
| *Encoders and amplifier* | | |
| | Encoder resolution (quadrature) | $0.00307\,\mathrm{rad/count}$ |
| | Recommended output voltage | $\pm 10\,\mathrm{V}$ |

**[TODO:** Keep only the rows for the module used; add any identified parameters (e.g. viscous friction) and the identification method.**]**

### Q1.3) System Model Discretisation

A sampling time of $T_s$ = **[TODO:**$T_s$**]** is selected because **[TODO:** justification, e.g. relative to the closed-loop bandwidth**]**. Using a zero-order hold (`c2d`), the discrete-time model is

$$x(k+1) = A\,x(k) + B\,u(k), \tag{2a}$$
$$y(k) = C\,x(k) + D\,u(k). \tag{2b}$$

**[TODO:** Numerical $A$, $B$, $C$, $D$.**]**

## Q2: State-Feedback Control (SFC) **(40 marks)**

### Q2.1) Controllability

For this question, the controllability matrix of the continuous-time model is

$$\mathcal{C} = \begin{bmatrix} B_c & A_cB_c & \cdots & A_c^{n-1}B_c \end{bmatrix}. \tag{3}$$

**[TODO:** Numerical $\mathcal{C}$ (`ctrb`), its rank and justification.**]**

### Q2.2) State and Input References (Equilibrium Point)

Given the desired output reference $y^\star$, the steady-state references $x^\star$ and $u^\star$ satisfy

$$\begin{bmatrix} A - I & B \\ C & D \end{bmatrix} \begin{bmatrix} x^\star \\ u^\star \end{bmatrix} = \begin{bmatrix} 0 \\ y^\star \end{bmatrix}. \tag{4}$$

**[TODO:** Solve for $x^\star$, $u^\star$ for your setup.**]**

### Q2.3) Closed-Loop Eigenvalues

The control requirements are an overshoot below $15\,\%$ and a settling time below $5\,\mathrm{s}$. The damping ratio and natural frequency follow from

$$\zeta = \frac{-\ln(OS/100)}{\sqrt{\pi^2 + \ln^2(OS/100)}}, \qquad \omega_n = \frac{4}{\zeta\, t_s}, \tag{5}$$

and the discrete-time eigenvalues from $z_i = e^{s_i T_s}$.

**[TODO:** Chosen $s_i$ and $z_i$ for all $n$ eigenvalues, with margin.**]**

### Q2.4) State-Feedback Gain

For the control law

$$u(k) = -F\big(x(k) - x^\star\big) + u^\star, \tag{6}$$

**[TODO:** $F$ from `place(A,B,pz)` and verification of $\mathrm{eig}(A - BF)$.**]**

### Q2.5) Closed-Loop Implementation

*[ Insert `assets/images/fig_q25_linear.png` here ]*

Figure 1: Closed-loop response of the linear model under state feedback.

**[TODO:** Comment on Figure 1: overshoot, settling time, steady-state error and control effort versus requirements. Add zooms.**]**

## Q3: Luenberger Observer Design **(20 marks)**

### Q3.1) Observability

The observability matrix of the continuous-time model is

$$\mathcal{O} = \begin{bmatrix} C_c \\ C_c A_c \\ \vdots \\ C_c A_c^{n-1} \end{bmatrix}. \tag{7}$$

**[TODO:** Numerical $\mathcal{O}$ (`obsv`), its rank and justification.**]**

### Q3.2) Observer Eigenvalues

**[TODO:** Observer eigenvalues (typically several times faster than the SFC eigenvalues) in continuous and discrete time.**]**

### Q3.3) Observer Gain

For the estimator

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + L\big(y(k) - \hat{y}(k)\big), \tag{8}$$

**[TODO:** $L$ from `place(A',C',po_z)'` and verification of $\mathrm{eig}(A - LC)$.**]**

### Q3.4) Closed-Loop Implementation

The controller now uses the estimated state:

$$u(k) = -F\big(\hat{x}(k) - x^\star\big) + u^\star. \tag{9}$$

*[ Insert `assets/images/fig_q34_estimation.png` here ]*

Figure 2: True and estimated states with $\hat{x}(0) \neq x(0)$.

*[ Insert `assets/images/fig_q34_closed_loop.png` here ]*

Figure 3: Closed-loop response with observer-based state feedback.

**[TODO:** Comment on Figures 2 and 3: observer convergence and closed-loop performance against requirements.**]**

## References

[1] R. P. Aguilera, *Lecture Notes on State-Feedback Control*, University of Technology Sydney, Australia, 2026.

[2] B. Messner and D. Tilbury, *Control Tutorials for MATLAB and Simulink*, University of Michigan, USA. [Online]. Available: http://ctms.engin.umich.edu/CTMS

[3] Quanser Inc., *QUBE-Servo 2 User Manual*, v1.0, Markham, ON, Canada.